

ω Cen: the mass-modelling procedure

oCen_dm project notes

2026 September 20

Abstract

The companion note describes how the kinematic data were prepared. This one describes what is done with them: the mass model, the dynamical solver, the likelihood, the priors, and the inference. The question is whether ω Cen requires mass beyond its stars, remnants and any central black hole, and it is posed as a comparison of Bayesian evidence between two model families that differ by one component. Everything here is deliberately low-dimensional: the data are 89 binned points over two decades in radius, and the failure mode to avoid is giving the halo more freedom than they can constrain.

1 The question, and how it is posed

Two families are fitted to the same observations and compared by evidence.

K1, no dark matter stars, a compact remnant population, and a central point mass.

K2, dark matter the same three components plus a truncated generalized NFW halo.

K1 is nested inside K2 at zero halo mass, so the Bayes factor $\ln Z_{K2} - \ln Z_{K1}$ measures whether the extra two parameters earn their prior volume. Two K2 variants are run separately, a cored halo ($\gamma = 0$) and a cuspy one ($\gamma = 1$), rather than fitting γ : making the inner slope free gives the halo shape freedom the data do not support, and the two limiting cases bracket it.

2 The mass model

The potential is spherical and the components add:

$$M(< r) = M_{\star}(< r) + M_{\text{rem}}(< r) + M_{\bullet} + M_{\text{DM}}(< r). \quad (1)$$

Stars. A multi-Gaussian expansion, deprojected analytically under spherical symmetry, scaled by a single free total mass $M_{\star}$. Using an MGE rather than a Plummer sphere matters because the tracer density appears in the Jeans equation directly, so its shape is not a free choice.

The default profile is not the surface brightness alone. It is a *composite*: HST star counts inside $25''$ spliced onto the Trager surface-brightness profile outside, because the light profile is unreliable in the crowded core where a handful of bright giants dominate it. The Trager-only alternative is selectable. The same shape supplies both the tracer density $\nu(r)$ and the stellar mass density, which assumes a radius-independent mass-to-light ratio; mass segregation makes that an approximation.

Remnants. A Plummer sphere with free mass M_{rem} and scale a_{rem} . It is a separate component rather than an inflation of the stellar mass-to-light ratio so that its mass appears under its own name in every posterior and can be confronted with multi-mass predictions later. The parametrisation is phenomenological: the point is to let a centrally concentrated dark component of *known* astrophysical origin compete with the halo, so that a halo detection cannot be manufactured by forbidding one.

Central point mass. $M_\bullet$, an intermediate-mass black hole or any unresolved central mass.

Dark halo (K2 only). A truncated generalized NFW profile,

$$\rho(r) = \rho_s \left(\frac{r}{r_s}\right)^{-\gamma} \left(1 + \frac{r}{r_s}\right)^{\gamma-3} T(r), \quad T(r) = \left[1 + \left(\frac{r}{r_t}\right)^2\right]^{-2}. \quad (2)$$

The truncation is smooth everywhere, keeps the total mass finite and falls as r^{-4} outside r_t , in the manner of a tidally limited halo. It is fixed at $r_t = 1000$ pc, well outside the data, which end at 57 pc.

The halo is parametrised by $M_{\text{DM}}(< 100 \text{ pc})$ and r_s rather than by ρ_s and r_s . This is not cosmetic: ρ_s and r_s are strongly degenerate and a prior flat in $\log \rho_s$ puts most of its mass where the halo is unobservable, whereas an enclosed mass at a radius near the data is close to what the data measure.

3 Anisotropy

The orbital anisotropy is the principal degeneracy with mass, so it is fitted, not assumed:

$$\beta(r) = \beta_0 + (\beta_\infty - \beta_0) \frac{r^2}{r^2 + r_\beta^2}. \quad (3)$$

Three parameters, monotonic, with closed-form $\exp(2 \int \beta/t \, dt)$, which is what the Jeans integrating factor needs. It is also exactly JamPy's logistic anisotropy with $\alpha = 2$, which is what makes the cross-check in §8 an identity rather than an approximation.

$\beta_0 \leq -1/2$ is imposed. [An & Evans \(2006\)](#) give two conditions. For a self-gravitating cusp, $\beta \leq \gamma/2$, so a cored tracer needs $\beta_0 \leq 0$. But in a potential dominated by a central point mass the condition is $\gamma \geq \beta + 1/2$, so a cored tracer needs $\beta_0 \leq -1/2$. Every model here carries a point mass, and its sphere of influence is *not* safely unresolved: it reaches the innermost datum at 0.11 pc once $M_\bullet > 7400 M_\odot$, which is 46 per cent of the prior. The tighter ceiling is applied to both families equally. It is a necessary, not sufficient, condition for a non-negative DF.

4 The dynamical solver

Given $\nu(r)$ from the tracer MGE, $M(< r)$ from the mass model and $\beta(r)$, the spherical Jeans equation

$$\frac{d(\nu\sigma_r^2)}{dr} + \frac{2\beta\nu\sigma_r^2}{r} = -\frac{\nu GM(< r)}{r^2} \quad (4)$$

integrates with the factor $g(r) = \exp(2 \int \beta/t \, dt)$ to

$$\nu\sigma_r^2(r) = \frac{1}{g(r)} \int_r^\infty g(s) \nu(s) \frac{GM(< s)}{s^2} ds, \quad (5)$$

and the projected second moments follow ([Binney & Mamon, 1982](#)):

$$\Sigma\sigma_{\text{los}}^2(R) = 2 \int_R^\infty \left[1 - \beta \frac{R^2}{r^2}\right] \nu\sigma_r^2 \frac{r \, dr}{\sqrt{r^2 - R^2}}, \quad (6)$$

$$\Sigma\sigma_{\text{pmR}}^2(R) = 2 \int_R^\infty \left[1 - \beta + \beta \frac{R^2}{r^2}\right] \nu\sigma_r^2 \frac{r \, dr}{\sqrt{r^2 - R^2}}, \quad (7)$$

$$\Sigma\sigma_{\text{pmT}}^2(R) = 2 \int_R^\infty [1 - \beta] \nu\sigma_r^2 \frac{r \, dr}{\sqrt{r^2 - R^2}}. \quad (8)$$

All three projections and $\Sigma(R)$ share their quadrature nodes, so one model evaluation delivers the line-of-sight, radial and tangential predictions together.

Numerics. The substitution $r = R \cosh u$ removes the integrable singularity at $r = R$ exactly, since $r dr / \sqrt{r^2 - R^2} = r du$, leaving smooth integrals that fixed Gauss–Legendre nodes handle to high accuracy. $\nu\sigma_r^2$ is tabulated once per model on a 600-point logarithmic grid and splined in the log. A full set of projected profiles costs a few milliseconds, which is what makes nested sampling over 10^5 – 10^6 likelihood calls practical.

One defect worth recording: the outer integral was originally evaluated as cumulative[−1] – cumulative, which loses all significance once the remaining integral is small, and σ_r collapsed to zero around three MGE scale lengths. It is now accumulated in reverse. The runs made before that fix are quarantined.

5 The likelihood

Each dataset is a set of binned dispersions with asymmetric errors. The log-likelihood is a split normal,

$$\ln \mathcal{L} = \sum_i -\frac{1}{2} \left(\frac{m_i - d_i}{\varepsilon_i} \right)^2 + \ln \frac{\sqrt{2/\pi}}{\varepsilon_i^- + \varepsilon_i^+}, \quad \varepsilon_i = \begin{cases} \varepsilon_i^+ & m_i > d_i \\ \varepsilon_i^- & \text{else,} \end{cases} \quad (9)$$

summed over datasets, which are treated as independent. The normalisation is *shared* between the two sides, which is what makes the density continuous at $m_i = d_i$ and integrate to one; giving each side its own $-\ln \varepsilon$, as this did until 2026-09-20, produced a jump of $\varepsilon^+/\varepsilon^-$ across every datum.

Three things happen between the model and the datum.

Radial averaging. For the datasets we measured ourselves — the four HST and *Gaia* profiles — the model is averaged over the radii of the stars actually measured in each bin, eight equal-count quantiles stored with the data, rather than over a complete annulus with $\Sigma(R)R$ weighting. Where the selection’s coverage changes inside a bin the two differ: HST’s 300–340'' sample has a star-weighted mean radius of 313.2'' against the annulus midpoint of 320'', worth up to 1.17σ . The 29 **MUSE** bins are published, carry no such information, and are still averaged over complete annuli.

Streaming. The model predicts the full second moment; the data are dispersions about a rotating mean. Each dataset that needs it carries $\langle \bar{v}^2 \rangle$, and the comparison is with $\sqrt{\sigma_{\text{model}}^2 - \langle \bar{v}^2 \rangle}$. The uncertainty on that term is propagated into the datum’s error for the HST and *Gaia* profiles. **It is not for MUSE**, whose rotation and dispersion were fitted jointly by the survey and would need their covariance rather than an independent error: at the innermost bin the first-order rotation contribution is about 1.22 km s^{-1} against quoted errors of $1.90/2.52 \text{ km s}^{-1}$, so this is not negligible and is an outstanding item.

Instrument scales. Optional multiplicative nuisances s_{inst} on the model prediction, one per instrument, with uniform priors. They absorb a constant calibration offset. They are also strongly degenerate with the halo mass, which is why the headline runs are done both with and without them: a halo that only survives when an instrument scale is free is not a detection.

6 Parameters and priors

7 Inference

Nested sampling with **ultranest**: 400 live points, $d \ln Z < 0.5$, MLFriends by default with a slice stepper available for the higher-dimensional variants. The evidence is the output of interest; the posterior comes with it.

Parameter	Prior	Unit	Meaning
$M_\star$	log-uniform $[10^6, 10^7]$	$M_\odot$	stellar mass
M_{rem}	log-uniform $[10^4, 3 \times 10^6]$	$M_\odot$	remnant mass
a_{rem}	log-uniform $[0.3, 20]$	pc	remnant scale radius
$M_\bullet$	log-uniform $[10^2, 3 \times 10^5]$	$M_\odot$	central point mass
β_0	uniform $[-1, -1/2]$	—	central anisotropy (An & Evans)
β_∞	uniform $[-1, 1]$	—	outer anisotropy
r_β	log-uniform $[0.5, 100]$	pc	anisotropy scale
$M_{\text{DM}}(< 100 \text{ pc})$	log-uniform $[10^3, 3 \times 10^7]$	$M_\odot$	K2 only
r_s	log-uniform $[3, 1000]$	pc	K2 only
s_{inst}	uniform, $\sim [0.7, 1.3]$	—	optional, one per instrument
D	normal (5.43, 0.05)	kpc	optional

Table 1: K1 has seven physical parameters, K2 nine. Log-uniform priors on every mass and scale, because they span orders of magnitude and the prior should not privilege a decade.

Run-to-run scatter in $\ln Z$ was measured by repetition at 0.5, so differences below about 1.5 are not interpretable. Each run writes the equally-weighted posterior, a summary with the evidence and per-dataset χ^2 at the maximum-likelihood sample, radial profiles for a random subset of samples, and a provenance record: git commit, sampler settings, priors, and the SHA-256 of every input product.

Evidences are comparable only between runs fitted to the *same observations*. The comparison tool fingerprints the dataset keys, the point count, the input hashes and the mock provenance, and refuses to quote a Bayes factor across a mismatch. Matching dataset names is not sufficient, because a product can be rebuilt under an unchanged key.

8 Cross-checks

The inference is not allowed to rest on one implementation.

A second Jeans engine. JamPy’s spherical solver runs behind the same likelihood. The mass components are expanded as projected MGEs so it can consume them, and the anisotropy family is chosen to be exactly JamPy’s logistic form, so agreement is an identity check on the implementation rather than an approximate comparison.

A distribution-function model. An AGAMA Cuddeford–Osipkov–Merritt DF, again behind the same likelihood. It is a *limited* cross-check and is described as such: a successful construction does not certify a non-negative distribution function, because AGAMA clips negative values to zero, which can change the realised tracer density (a ratio of 1.045 at 0.11 pc in one trial). Two further limits: its Cuddeford family cannot represent the anisotropy profile we fit, and the backend softens the point mass with a 10^{-4} pc Plummer radius, so its central limit is not Keplerian. It is run as a diagnostic on posterior draws, not as a veto — vetoing would define a different model and require its own evidence.

Published analyses. Presets reproduce the assumptions of individual papers and compare the numbers we recover with theirs, as regression anchors.

Injection. Mock data generated from a K1 model and fitted with both families, to confirm the evidence comparison does not manufacture a halo when there is none.

9 What the model does not include

Stated plainly, because each is a route by which a spurious halo could appear.

1. **Sphericity.** ω Cen is flattened. A spherical model absorbs flattening into mass and anisotropy.
2. **Rotation is removed, not modelled.** The streaming term is subtracted from the model's second moment. A rotating model would be preferable.
3. **Equilibrium.** The cluster is tidally limited and losing stars; the data end at $0.7 r_J(\text{peri})$, and the outermost bins are closest to where equilibrium is least safe.
4. **Diagonal likelihood.** The error-model choice, the rotation-curve uncertainty and the shared field template all act coherently across bins but enter as independent per-bin errors. A shared nuisance parameter is the correct treatment; comparing separate fits is the chosen alternative.
5. **A single tracer population.** Mass segregation makes the dispersion depend on stellar mass, and the selections differ in magnitude between instruments.

10 The runs

They are taken one rung at a time, following `docs/MODELLING_PLAN.md`: constant anisotropy without instrument scales first (6 and 8 parameters against 89 points), then the three-parameter $\beta(r)$, then the scales. The decision to climb is made on the residuals of the rung below, not in advance. An anisotropy turnover and a shared rotation nuisance are written down there as deferred, with the conditions that would bring them back.

The default dataset is the 89 points described in the companion note. Two modelling choices are left explicit as switches rather than averaged over, and the intention is to fit each and compare rather than to guess:

Run	Command	Purpose
K1	<code>-family K1</code>	no dark matter, the null
K2 cored	<code>-family K2-cored</code>	the headline comparison
K2 NFW	<code>-family K2-nfw</code>	the cuspy bracket
inflated errors	<code>-gaia-errors eta</code>	sensitivity to the Gaia error model
our rotation	<code>-gaia-rotation ours</code>	sensitivity to the rotation curve
no scales	<code>-no-scales</code>	does the halo survive without nuisances
injection	<code>-mock-from ...</code>	does the comparison produce a false positive

References

- An, J. H., Evans, N. W. 2006, ApJ, 642, 752. <https://ui.adsabs.harvard.edu/abs/2006ApJ...642..752A>
- Binney, J., Mamon, G. A. 1982, MNRAS, 200, 361. <https://ui.adsabs.harvard.edu/abs/1982MNRAS.200..361B>
- Vasiliev, E., Baumgardt, H. 2021, MNRAS, 505, 5978. <https://arxiv.org/abs/2102.09568>